arXiv:2406.09211v1 [cs.CV] 13 Jun 2024

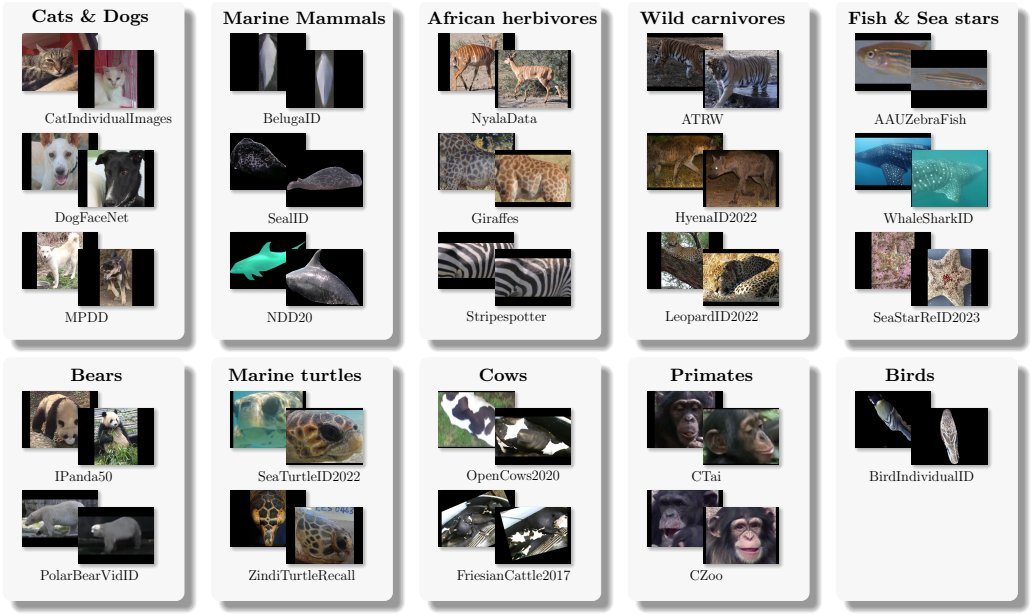
WildlifeReID-10k: Wildlife re-identification dataset with 10k individual animals

Lukáš Adam[1*], **Vojtěch Čermák**[2], **Kostas Papafitsoros**[3], **and Lukas Picek**[1,4]

[1] University of West Bohemia, [2] Czech Technical University in Prague,
[3] Queen Mary University of London, and [4] INRIA.

Abstract

We introduce a new wildlife re-identification dataset WildlifeReID-10k with more than 214k images of 10k individual animals. It is a collection of 30 existing wildlife re-identification datasets with additional processing steps. WildlifeReID-10k contains animals as diverse as marine turtles, primates, birds, African herbivores, marine mammals and domestic animals. Due to the ubiquity of similar images in datasets, we argue that the standard (random) splits into training and testing sets are inadequate for wildlife re-identification and propose a new similarity-aware split based on the similarity of extracted features. To promote fair method comparison, we include similarity-aware splits both for closed-set and open-set settings, use MegaDescriptor - a foundational model for wildlife re-identification - for baseline performance and host a leaderboard with the best results. We publicly publish the dataset and the codes used to create it in the `wildlife-datasets` library, making WildlifeReID-10k both highly curated and easy to use.



[*] Dataset: `https://www.kaggle.com/datasets/wildlifedatasets/wildlifereid-10k`
Leaderboard: `https://paperswithcode.com/dataset/wildlifereid-10k`

Preprint. Under review.

1 Introduction

Table 1: WildlifeReID-10k summary.

	images	individuals
birds	52,274	50
cats	13,021	509
chimpanzees	6,771	95
cows	20,724	375
dogs	10,020	1,584
dolphins	2,657	82
fish	6,672	6
giraffes	2,032	329
hyenas	3,129	256
leopards	6,806	430
nyalas	1,942	237
pandas	6,874	50
polar bears	13,918	13
sea stars	2,187	95
sea turtles	21,532	2,703
seals	2,080	57
sharks	7,693	543
tigers	5,415	182
whales	8,559	788
zebras	19,956	1,960
TOTAL	214,262	10,344

Wildlife re-identification involves assigning identities (classes) to individual wild animals in images. This process typically examines morphological characteristics of animals, such as patterns, which are stable over time and unique to each individual. When combined with other image metadata (e.g. capture time and date, location, activity, behaviour), this information is highly beneficial for studying various aspects of wild animal populations' biology and ecology. It can be used for tasks such as disease monitoring and control [31], understanding the animal's role in the ecosystem [36], monitoring invasive species [10], or measuring the involvement of humans in the animal's habitat and ecological restoration [8].

In recent years, there has been a significant increase in research on automatic re-identification methods for wild animals. This surge has been made possible by recent advances in computer vision and machine learning and has also been driven by the growing availability of wildlife imagery obtained through various sources such as camera traps [38], drones [23], satellites [46], regular photo-identification surveys [39], and passive crowdsourcing from social media data [32]. This increase in research is evidenced not only by the increasing number of research studies [7, 9, 35] and the corresponding review papers [37, 44, 34, 38] but also by the expanding number of publicly available datasets for wildlife re-identification. These datasets cover several animal groups, including primates [16, 48], carnivores [24, 42, 12], reptiles [2, 13], whales [11], or mammals [1, 41, 50].

However, the growth of this field is also accompanied by several limitations of the published datasets. Due to labelling costs, they often contain only limited numbers of photos and individuals, and therefore, have a limited standalone value. The datasets often lack documentation or a unified format, making their initial analysis time-consuming. Even worse, they typically do not have standardised training-testing splits and evaluation metrics. Consequently, newly proposed algorithms usually evaluate their performance only on a fraction of available datasets and do not compare their performance with prior work [21, 37, 40]. In our previous work [43], we addressed some of these issues by releasing the Python library `wildlife-datasets` which greatly simplifies dataset handling, and by training MegaDescriptor, a foundational model for wildlife re-identification.

In this paper, we build on this work to provide a reliable benchmark that reflects the specifics of animal datasets. Wildlife re-identification datasets often contain multiple similar photos that have been taken during a single human-animal encounter or extracted from subsequent video frames. In Figure 1 we illustrate an example of this situation with turtle "t405", which appears in 15 images of the SeaTurtleID2022 dataset. By clustering DINOv2 features, we identified 3 clusters of similar images (yellow, blue, and brown). When the images of this individual are split randomly, images from the blue cluster end up both in the training and testing set, making their identification trivial - similarly for the brown cluster. This results in an overestimation of the performance of the tested re-identification method. To prevent this undesired behaviour, we propose a new *similarity-aware split*. We include *all clusters* in the training set and then split the unclustered images into the training and testing sets randomly. As seen in Figure 1, this means that the two unclustered images are in the testing set, making their identification much more challenging with respect to the random split. We see this similarity-aware split as an extension of the time-aware split [3], which was based on a cutoff time, with all images taken before (after) a certain date assigned to the training (testing) set. However, since most datasets do not provide timestamps, the time-aware split cannot usually be created. On the other hand, the proposed similarity-aware split can be implemented in all cases.

To create a reliable benchmark for wildlife re-identification, we applied our processing pipeline [43] to 30 wildlife re-identification datasets, which resulted in the creation of WildlifeReID-10k, a large dataset with 214k images of 10k individual animals. Compared to our previous work in [43], we

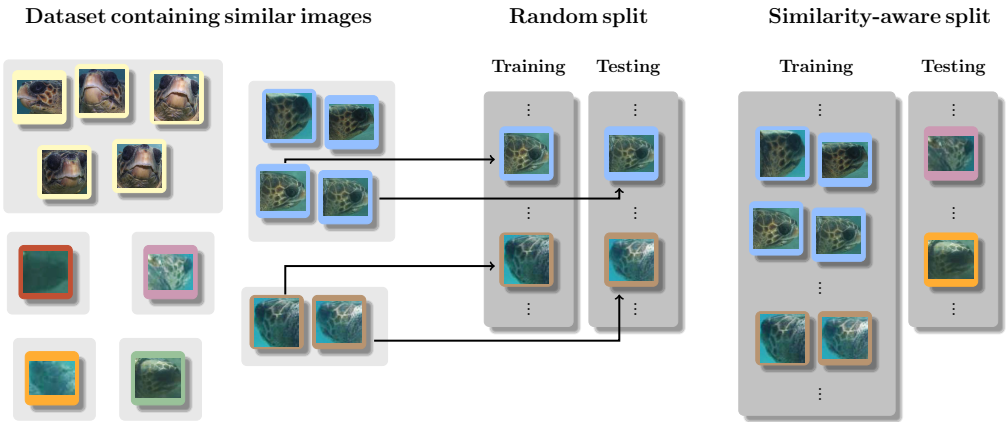


Figure 1: Left: Dataset containing similar images (clusters denoted by different colours), taken during a single encounter. Middle: Random split places similar images into the training and testing set resulting in performance overestimation for the re-identification method. Right: Proposed similarity-aware split assigns all the clusters to the training set, mitigating this undesirable information leakage.

performed multiple enhancements: (i) 11 datasets are completely new or have updated images, (ii) we improved the collection quality by excluding 2 poor datasets and reducing 2 more datasets to a tenth of their original size, (iii) we reduced the dataset size from 117GB to 23GB by removing unneeded files (videos) and applying bounding boxes, (iv) by uploading the dataset on Kaggle, we prevented issues with some datasets that previously would need to be handled manually. Additionally, to facilitate future research, we also include the similarity-aware split for both closed-set and open-set settings, provided baseline results obtained with MegaDescriptor [43] and a leaderboard.

2 Related work

Even though there are several review papers concerning wildlife re-identification methods [34, 37, 38, 44], we are unaware of any paper performing large-scale benchmarking across a variety of datasets.

An overview of public re-identification datasets was recently published in [43]. A Python library was created for handling the datasets and was used to train a feature-extracting model MegaDescriptor. It was further shown that for wildlife re-identification, MegaDescriptor outperforms both general feature-extracting deep methods and classical feature matching algorithms.

The authors in [21] created the dataset Wildlife-71. They used three wildlife re-identification datasets ATRW, GiraffeZebraID, SealID (compare with 30 datasets in WildlifeReID-10k in Table 2) and manually labelled 1,016 identities from the GOT-10k tracking dataset and 816 animal Youtube videos. Therefore, a significant portion of the dataset is made from video datasets, reducing the diversity of the images. Besides having potential license issues, at the moment of writing, this dataset has not been made public yet.

3 WildlifeReID-10k

3.1 Dataset creation

This work builds on `wildlife-datasets` [43] – that unifies available wildlife re-identification datasets and provides an API that allows their straightforward access and download. In the first release, the WildlifeDatasets contained 30 datasets. Since then we have added 6 new datasets (CatIndividualImages, CowDataset, DogFaceNet, MPDD, PolarBearVidID, SeaStarReID2023). From these 36 datasets, we were unable to include 4 datasets due to their license files and decided to exclude 2 additional datasets due to their poor quality. All the remaining datasets have permissible licenses for being re-uploaded.

We provide a basic summary of used datasets in Table 2. The dataset contains 214,262 images of 10,344 individuals. All images were either already cropped or we applied the provided bounding

boxes or segmentation masks. This reduced the size of the dataset to only 23.06GB or equivalently 107kB per image on average. We did not resize any images.

To improve dataset quality, we performed additional modifications to some datasets. To prevent repeating similar images, we took only every tenth frame from datasets created from videos with high FPS (AerialCattle2017 and PolarBearVidID). We updated 2 datasets which published new data (BelugaID, SeaTurtleID2022) and fixed several wrong labels in 3 additional datasets (CTai, Cows2021, FriesianCattle2015). In all datasets we also found unidentified animals (being named "Adult", "new_whale" or "____" in different datasets) and excluded them from the final dataset. Therefore, more than one third of the constituting datasets in WildlifeReID-10k are either brand new or contain additional images with respect to [43]. Detailed scripts used to create WildlifeReID-10k are available within the `wildlife-datasets` library.

Table 1 shows the species distribution in WildlifeReID-10k. While there are several domestic animals, most of the animals are wild. Indeed, Table 2 implies that 50% individuals were observed in the wild (72% when including rehabilitation centres). This number excludes wild animals in the zoos and confirms that the WildlifeReID-10k dataset indeed focuses mostly on wild animals.

Table 2: Datasets used to create WildlifeReID-10k.

	images	individuals	size (GB)	wild	license
AAUZebraFish [9]	6,672	6	0.37	✗	CC BY 4.0
AerialCattle2017 [5]	4,700	23	0.03	✗	NC-Government
ATRW [24]	5,415	182	0.27	✗	CC BY-NC-SA 4.0
BelugaID [1]	8,559	788	0.26	✓	CDLA-Permissive-1.0
BirdIndividualID [15]	52,274	50	0.50	✗	None
CatIndividualImages [25]	13,021	509	9.01	✗	CC BY 4.0
CowDataset [17]	1,485	13	1.66	✗	CC BY 4.0
Cows2021 [18]	8,670	179	0.11	✗	NC-Government
CTai [16]	4,662	71	0.33	✓	Other
CZoo [16]	2,109	24	0.24	✗	Other
DogFaceNet [28]	8,363	1,393	0.08	✗	MIT
FriesianCattle2015 [6]	193	25	0.03	✗	NC-Government
FriesianCattle2017 [5]	940	89	0.18	✗	NC-Government
Giraffes [27]	1,393	178	0.09	✓	None
GiraffeZebraID [33]	6,925	2,056	1.03	✓	CDLA-Permissive-1.0
HyenaID2022 [42]	3,129	256	0.44	✓	CDLA-Permissive-1.0
IPanda50 [47]	6,874	50	0.35	✗	None
LeopardID2022 [42]	6,806	430	0.85	✓	CDLA-Permissive-1.0
MPDD [19]	1,657	191	0.02	✗	CC BY 4.0
NDD20 [41]	2,657	82	0.07	✓	CC BY-NC-SA 4.0
NyalaData [12]	1,942	237	0.46	✓	None
OpenCows2020 [4]	4,736	46	0.04	✗	NC-Government
PolarBearVidID [49]	13,918	13	0.07	✗	CC BY 4.0
SealID [29]	2,080	57	0.26	✓	CC BY 4.0
SeaStarReID2023 [45]	2,187	95	1.66	✗	CDLA-Permissive-1.0
SeaTurtleID2022 [3]	8,729	438	0.13	✓	Other
SMALST [50]	12,850	10	0.84	✗	MIT
StripeSpotter [22]	820	45	0.02	✓	CC BY-SA 3.0
WhaleSharkID [20]	7,693	543	1.57	✓	CDLA-Permissive-1.0
ZindiTurtleRecall [2]	12,803	2,265	2.09	✗	CC BY-SA 4.0
TOTAL	214,262	10,344	23.06		

3.2 Dataset availability

The WildlifeReID-10k dataset is publicly available under the license that all its comprising datasets need to be cited and that all their licenses must be satisfied. Therefore, it is prohibited to use it for commercial research. We intend to add any dataset added to the `wildlife-datasets` library, to

WildlifeReID-10k as well (if permitted by license). Since both splits and leaderboard are applied dataset-wise, adding new datasets will interfere with neither of them.

4 How to create similarity-aware splits

One major drawback of wildlife re-identification datasets is the inconsistent methodology for training-testing splits. Firstly, the closed-set setting, commonly used in machine learning, requires all individuals (classes) in the testing set to be in the training set. This setting, however, cannot capture evolving populations, which is a standard setting in living systems. Secondly, developers of re-identification algorithms usually assign images to the training and testing sets randomly. However, wildlife re-identification datasets typically contain similar images such as when multiple photos are taken during a single human-animal encounter or when subsequent video frames are extracted from a video. Placing images into the training and testing set randomly results in situations when similar images are both in the training and testing set and hence, the classification of such images is trivial.

To the best of our knowledge, all of the 30 datasets from Table 2 either provide no default split or a random split without considering information about data acquisition. The only exception is the SeaTurtleID2022 dataset, where default splits based on times were proposed. Images taken before a date are assigned to the training set and images taken after a date are assigned to the testing set. It was shown that not considering this split leads to a significant training-to-testing leak and an artificially inflated performance reaching tens of per cent.

Here we propose a general method for generating splits mitigating the training-to-testing information leakage. Naturally, we provide its full implementation in the `wildlife-dataset` library. Before we outline the details of our method, we first briefly describe the closed-set and open-set splits, which are two general split categories describing whether new identities appear during inference.

4.1 Splits based on identities in the training and testing sets

The **closed-set setting** refers to the scenario where there are some previously identified individuals and all new images show these individuals. The usual goal is to assign one of the known identities to each image with an unidentified individual. The term closed-set follows from the fact that the population is fully known and no additional individuals may appear. This setting is suitable for cases with full control over the observation environment, such as animals in a zoo or livestock on a farm.

The **open-set setting** refers to the scenario where there are some previously identified individuals, but new images may contain previously unknown individuals. Examples of these individuals include newly born or previously unobserved animals. The usual goal is to assign one of the known identities to each image or to state that it is a new individual. The term open-set follows from the fact that the population may change. This setting is suitable for cases where there is a known part of the population, but new individuals may still appear and need to be recognized. This scenario is common in biology applications, therefore, open-set splits are preferable for wildlife re-identification.

4.2 Similarity-aware splits

The simplest strategy to prevent information leakage is to use some external markers such as timestamps. However, this information is often not available. Here we propose a simple procedure based on clustering features. Our algorithm requires an initial split and it keeps the same number of images in both the training and testing sets for each individual as the initial split. Therefore, if the original split is open-set, the new split will be open-set as well. This allows to generate splits with a priori specified properties.

We describe the whole procedure in Algorithm 1. The split is computed for each individual separately, As we have already mentioned, for each individual, we require $n_{\text{train}}^{\text{individual}}$ to be its number of images in the training set. We extract the DINOv2 features [30] and use DBSCAN [14] as the clustering algorithm. DBSCAN has two parameters; we use fixed $\min_{\text{samples}} = 2$ and varying ε. The former parameter allows for clusters of at least two images. The latter parameter ε determines the cluster radius, where all points are outliers for $\varepsilon = 0$ and all points form one cluster for $\varepsilon = \infty$. We therefore start with a small ε and keep increasing it until all clusters consist of at most n_{cluster} images. Then we place all clustered images to the training set. Since clustered images may be considered as various

representations of the same image, this maximizes the variability between the training and testing sets.

We observed that the maximal cluster size $n_{\text{cluster}} = 5$ results in clusters containing enough individuals but not too many. We conjecture that this is connected with the specific data acquisition and handling strategies in wildlife re-identification when the researchers manually filter images upon capturing multiple photos during an encounter.

Algorithm 1 Split mitigating the training-to-testing leakage of similar images

Require: $n_{\text{train}}^{\text{individual}}$ number of images in the training size for each individual
for individual **in** dataset **do**
  for increasing $\varepsilon \in [\varepsilon_0, \varepsilon_1]$ **do**
    clusters $\leftarrow$ DBSCAN(ε, $\min_{\text{samples}} = 2$) on features of images of individual
    if any cluster size $> n_{\text{cluster}}$ **then break**
  end for
  take the previous ε with clusters of size $\leq n_{\text{cluster}}$
  put clusters into the training set while its size $\leq n_{\text{train}}^{\text{individual}}$
  distribute the remaining images randomly to fill the training set to size $= n_{\text{train}}^{\text{individual}}$
end for

Figure 2 shows 3 clusters (rows) found by Algorithm 1 in the LeopardID2022 dataset. The images are almost identical but differ in small details such as image size or the leopard's head position.



Figure 2: Clusters (rows) found by Algorithm 1 in the LeopardID2022 dataset.

5 Evaluation metrics

5.1 Used splits

We provide a default similarity-aware open-set split within WildlifeReID-10k. This split is intended for evaluation of both the open-set and closed-set settings. For the former, the classifier is supposed to predict either one of the 10,344 known identities or a new individual. For the latter, the new individuals in the testing set are ignored and the classifier predicts only one of the known identities. We provide a summary of both splits in Table 3.

Table 3: Provided default similarity-aware splits for WildlifeReID-10k.

	images			individuals		
	training	testing: known	testing: new	training	testing: known	testing: new
Closed-set	168,739	21,492	0	9,437	8,496	0
Open-set	168,739	21,492	24,031	9,437	8,496	907

5.2 Evaluation metrics

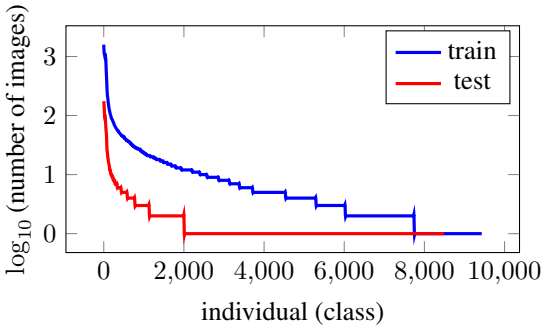


Figure 3: Number of images per individual in the training and testing sets in WildlifeReID-10k.

Figure 3 shows the number of images per individual in both the training and testing sets. We applied a logarithmic scale to the y-axis. There are big discrepancies in the testing set: Approximately 79% of individuals contain only one image in the testing set, while the top 1% of individuals amount for 34% of images and the top 10% of individuals amount for 58% of images. This imbalance is typical for datasets with images taken in the wild, where some animals might prefer the "popular" places (routes with camera traps, shallow waters close to the shore, ...) and subsequently have lots of images. The standard accuracy metric is therefore inadequate because it gives large weights to often-observed animals, which are often well-studied. Instead, we use the balanced accuracy.

We follow [26] and define BAKS (balanced accuracy on known samples) and BAUS (balanced accuracy on unknown samples) by

$$\begin{aligned} \text{BAKS} &= \frac{1}{D}\sum_{d=1}^{D}\frac{1}{C_d}\sum_{c=1}^{C_d}\frac{1}{|I_{\text{known}}(d,c)|}\sum_{i\in I_{\text{known}}(d,c)}\mathbf{1}(y_i=\hat{y}_i) \\ \text{BAUS} &= \frac{1}{D}\sum_{d=1}^{D}\frac{1}{|I_{\text{unknown}}(d)|}\sum_{i\in I_{\text{unknown}}(d)}\mathbf{1}(y_i=\hat{y}_i) \end{aligned} \tag{1}$$

Here, $\mathbf{1}$ is the indicator (0/1) function, D the number of datasets, C_d the number of known classes (individuals) in dataset d and $I_{\text{known}}(d,c)$ indices of all images of class c in the testing set of the dataset d. Finally, y_i and $\hat{y}_i$ are the true and predicted identities respectively of the i-th image. In other words, in BAKS, we compute three means, the outer with respect to datasets, the middle with respect to classes in the dataset and the inner with respect to images in the class. In BAUS, the middle mean is unnecessary because there is only one unknown class.

For the closed-set setting, we use BAKS as the evaluation metric. For the open-set setting, we use the normalized balanced accuracy, which is defined as the mean of BAKS and BAUS.

6 Baseline performance

To obtain a baseline performance, we train MegaDescriptor-L with the same hyperparameters as described in the original paper [43]. We trained two models in total, one for the random split and one for the similarity-aware split. The whole training lasted approximately one week on a single GPU. Since the training set is the same for both the closed-set and open-set settings, we used one trained model for both. After being trained, the last layer of MegaDescriptor is removed and it serves as a feature-extracting model. For the closed-set setting, inference is performed by finding the closest features (measured by the cosine similarity) between the testing image and all features of the training images. For the open-set setting, we inferred that an image depicts a new individual whenever the closest similarity score was below a threshold. For verification of the good performance of MegaDescriptor, we used DINOv2 features [30] with an identical inference procedure.

Figure 4 shows BAKS for the closed-set setting on the similarity-aware split for all datasets (here we slightly abuse the terminology since BAKS from (1) is the mean over all datasets). MegaDescriptor (blue) significantly outperforms DINOv2 (red) features. It is interesting to note that 12 datasets with the best performance are taken in controlled environments, while 11 datasets with the worst performance are from the wild (except for ZindiTurtleRecall which includes wild animals but in a rehabilitation centre). This shows the importance of wild datasets over datasets from controlled environments.

The procedure in Algorithm 1 generated clusters that consist of 35,259 images (16.5% of the whole dataset). In Table 4 we focus on the clusters in datasets whose images contained timestamps. The

left column shows the total number of clustered images. The right column shows the number of clustered images whose clusters contain only images taken during the same day. These images are typically taken during the same single encounter and are thus expected to be similar. In total, 76.1% (5458/7172) images formed clusters taken on the same day, validating the close resemblance between our proposed similarity-aware and the time-aware split [3]. The latter split keeps images taken on the same day either in the training or in the testing set and was designed to prevent evaluating re-identification methods on easier and non-realistic scenarios. The similarity-aware split is designed with the same purpose but can also be used in datasets without timestamps.

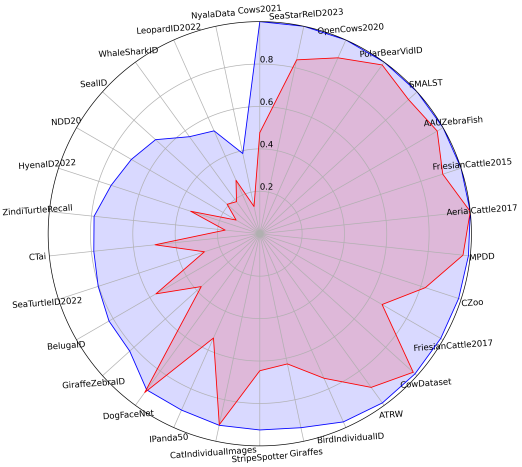


Figure 4: BAKS on individual datasets for MegaDescriptor (blue) and DINOv2 (red) features.

Table 4: Images in clusters produced by Algorithm 1 for datasets with timestamps. Total number of clustered images (left) and clustered images from the same day (right).

	animals in clusters	
	all	same day
BelugaID	2,125	2,036
Cows2021	1,458	942
GiraffeZebraID	841	724
SeaTurtleID2022	2,384	1,509
StripeSpotter	200	200
WhaleSharkID	164	47
TOTAL	7,172	5,458

To validate the last claim, in Figure 5 we investigate the impact of the similarity-aware split. The left subfigure shows BAKS for all datasets for MegaDescriptor (blue) and DINOv2 (orange) on similarity-aware and random splits. It confirms again that MegaDescriptor outperforms DINOv2. Furthermore, it shows that DINOv2 has a significantly better performance on the random split than on the similarity-aware split (on average by 3.7% on non-trivial datasets where at least one BAKS is below 99%). This difference decreases for MegaDescriptor (on average by 0.6% on non-trivial datasets) but is still present. The right subfigure shows BAKS and BAUS for various thresholding parameters described above. The interpretation is the same: while there is a significant difference for DINOv2 between the random (AUC 48.3%) and the similarity-aware (AUC 43.0%) split, for MegaDescriptor the difference between the random (AUC 77.0%) and the similarity-aware (AUC 76.7%) splits is smaller. This confirms that the random split removes complexity from the re-identification problem both when the model was trained on the data (MegaDescriptor) and when it never saw them (DINOv2). Therefore, dataset curators and model developers should consider using the similarity-aware split instead of the random split.

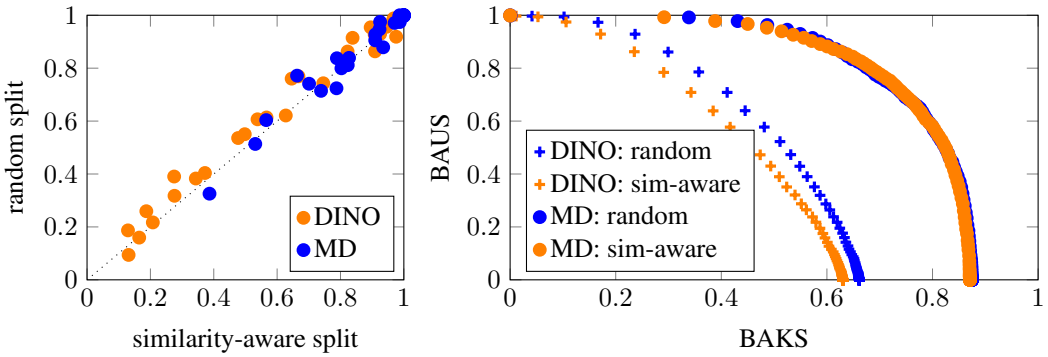


Figure 5: Performance difference for DINOv2 and MegaDescriptor between the random and similarity-aware splits for the closed-set (left) and open-set (right) settings.

7 Limitations and negative societal impact

Since WildlifeReID-10k contains images of animals and not humans and these images were already part of existing datasets, we are unaware of any negative societal impact. Regarding limitations, most biologists and their algorithm development partners are interested only in the species they study and do not need a general model able to perform well for many different animals. In such a case, it could be beneficial to use a specific dataset of the studied species instead of the general WildlifeReID-10k for method development and evaluation. Unfortunately, such datasets are seldom large enough to allow training from scratch, with the possible exception of HappyWhale [11]. However, we anticipate that using a general model based on WildlifeReID-10k, possibly fine-tuned on a small dataset of the focal species, may provide lots of benefits. Second, even though we provided compelling evidence, there are no theoretical guarantees that the similarity-aware split will always eliminate training-to-testing leakage. However, even in the worst case when the clusters from Algorithm 1 are poor (random), the similarity-aware split reduces to the random split.

8 Conclusions

We have introduced WildlifeReID-10k, a wildlife re-identification dataset of more than 214k images of 10k individual animals. To our knowledge, this is the largest publicly available collection of animal images, carefully curated and gathered in one place. To make WildlifeReID-10k as accessible as possible, we also made publicly available the codes to create, analyze and evaluate it.

We argued that splits in wildlife re-identification literature are usually missing or done randomly. We have shown that the latter leads to the training-to-testing leakage and makes the re-identification task more trivial, artificially inflating the performance of methods. To address that, we have proposed a new more realistic similarity-aware split and incorporated it into WildlifeReID-10k as a default split.

We anticipate that WildlifeReID-10k will be widely used by both computer vision scientists and ecologists interested in wildlife re-identification and will significantly facilitate progress in this field. Indeed, new algorithms will be easily tested in a standardised way, on more than a few datasets in contrast to the so far standard practices.

References

[1] Beluga ID 2022, 2022.

[2] Turtle recall: Conservation challenge, 2022.

[3] L. Adam, V. Čermák, K. Papafitsoros, and L. Picek. SeaTurtleID2022: A long-span dataset for reliable sea turtle re-identification. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 7146–7156, January 2024.

[4] W. Andrew, J. Gao, S. Mullan, N. Campbell, A. W. Dowsey, and T. Burghardt. Visual identification of individual holstein-friesian cattle via deep metric learning. *Computers and Electronics in Agriculture*, 185:106133, 2021.

[5] W. Andrew, C. Greatwood, and T. Burghardt. Visual localisation and individual identification of holstein friesian cattle via deep learning. In *Proceedings of the IEEE International Conference on Computer Vision Workshops*, pages 2850–2859, 2017.

[6] W. Andrew, S. Hannuna, N. Campbell, and T. Burghardt. Automatic individual holstein friesian cattle identification via selective local coat pattern matching in RGB-D imagery. In *2016 IEEE International Conference on Image Processing (ICIP)*, pages 484–488. IEEE, 2016.

[7] A. Bedetti, C. Greyling, B. Paul, J. Blondeau, A. Clark, H. Malin, J. Horne, R. Makukule, J. Wilmot, T. Eggeling, et al. System for elephant ear-pattern knowledge (SEEK) to identify individual african elephants. *Pachyderm*, 61:63–77, 2020.

[8] J. D. Blount, M. W. Chynoweth, A. M. Green, and Ç. H. Şekercioğlu. COVID-19 highlights the importance of camera traps for wildlife conservation research and management. *Biological Conservation*, 256:108984, 2021.

[9] J. Bruslund Haurum, A. Karpova, M. Pedersen, S. Hein Bengtson, and T. B. Moeslund. Re-identification of zebrafish using metric learning. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision Workshops*, pages 1–11, 2020.

[10] A. Caravaggi, M. Zaccaroni, F. Riga, S. C. Schai-Braun, J. T. Dick, W. I. Montgomery, and N. Reid. An invasive-native mammalian species replacement process captured by camera trap survey random encounter models. *Remote Sensing in Ecology and Conservation*, 2(1):45–58, 2016.

[11] T. Cheeseman, K. Southerland, J. Park, M. Olio, K. Flynn, J. Calambokidis, L. Jones, C. Garrigue, A. Frisch Jordan, A. Howard, et al. Advanced image recognition: A fully automated, high-accuracy photo-identification matching system for humpback whales. *Mammalian Biology*, 102(3):915–929, 2022.

[12] N. Dlamini and T. L. van Zyl. Automated identification of individuals in wildlife population using siamese neural networks. In *2020 7th International Conference on Soft Computing & Machine Intelligence (ISCMI)*, pages 224–228. IEEE, 2020.

[13] S. G. Dunbar, E. C. Anger, J. R. Parham, C. Kingen, M. K. Wright, C. T. Hayes, S. Safi, J. Holmberg, L. Salinas, and D. S. Baumbach. Hotspotter: Using a computer-driven photo-id application to identify sea turtles. *Journal of Experimental Marine Biology and Ecology*, 535:151490, 2021.

[14] M. Ester, H.-P. Kriegel, J. Sander, X. Xu, et al. A density-based algorithm for discovering clusters in large spatial databases with noise. In *KDD*, volume 96, pages 226–231, 1996.

[15] A. C. Ferreira, L. R. Silva, F. Renna, H. B. Brandl, J. P. Renoult, D. R. Farine, R. Covas, and C. Doutrelant. Deep learning-based methods for individual recognition in small birds. *Methods in Ecology and Evolution*, 11(9):1072–1085, 2020.

[16] A. Freytag, E. Rodner, M. Simon, A. Loos, H. S. Kühl, and J. Denzler. Chimpanzee faces in the wild: Log-Euclidean CNNs for predicting identities and attributes of primates. In *German Conference on Pattern Recognition*, pages 51–63. Springer, 2016.

[17] L. Fu and G. He. Cow dataset, 2021.

[18] J. Gao, T. Burghardt, W. Andrew, A. W. Dowsey, and N. W. Campbell. Towards self-supervision for video identification of individual holstein-friesian cattle: The Cows2021 dataset. *arXiv preprint arXiv:2105.01938*, 2021.

[19] Z. He, J. Qian, D. Yan, C. Wang, and Y. Xin. Animal re-identification algorithm for posture diversity. In *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–5. IEEE, 2023.

[20] J. Holmberg, B. Norman, and Z. Arzoumanian. Estimating population size, structure, and residency time for whale sharks Rhincodon typus through collaborative photo-identification. *Endangered Species Research*, 7(1):39–53, 2009.

[21] B. Jiao, L. Liu, L. Gao, R. Wu, G. Lin, P. Wang, and Y. Zhang. Toward re-identifying any animal. *Advances in Neural Information Processing Systems*, 36, 2024.

[22] M. Lahiri, C. Tantipathananandh, R. Warungu, D. I. Rubenstein, and T. Y. Berger-Wolf. Biometric animal databases from field photographs: identification of individual zebra in the wild. In *Proceedings of the 1st ACM international conference on multimedia retrieval*, pages 1–8, 2011.

[23] S. S. Landeo-Yauri, E. A. Ramos, D. N. Castelblanco-Martínez, C. A. Niño-Torres, and L. Searle. Using small drones to photo-identify antillean manatees: A novel method for monitoring an endangered marine mammal in the caribbean sea. *Endangered Species Research*, 41:79–90, 2020.

[24] S. Li, J. Li, H. Tang, R. Qian, and W. Lin. ATRW: A benchmark for Amur tiger re-identification in the wild. In *Proceedings of the 28th ACM International Conference on Multimedia*, page 2590–2598. Association for Computing Machinery, 2020.

[25] T.-Y. Lin. Cat individual images, 2020.

[26] P. R. Mendes Júnior, R. M. De Souza, R. d. O. Werneck, B. V. Stein, D. V. Pazinato, W. R. de Almeida, O. A. Penatti, R. d. S. Torres, and A. Rocha. Nearest neighbors distance ratio open-set classifier. *Machine Learning*, 106(3):359–386, 2017.

[27] V. Miele, G. Dussert, B. Spataro, S. Chamaillé-Jammes, D. Allainé, and C. Bonenfant. Revisiting animal photo-identification using deep metric learning and network analysis. *Methods in Ecology and Evolution*, 12(5):863–873, 2021.

[28] G. Mougeot, D. Li, and S. Jia. A deep learning approach for dog face verification and recognition. In *PRICAI 2019: Trends in Artificial Intelligence: 16th Pacific Rim International Conference on Artificial Intelligence, Cuvu, Yanuca Island, Fiji, August 26-30, 2019, Proceedings, Part III 16*, pages 418–430. Springer, 2019.

[29] E. Nepovinnykh, T. Eerola, V. Biard, P. Mutka, M. Niemi, M. Kunnasranta, and H. Kälviäinen. SealID: Saimaa ringed seal re-identification dataset. *Sensors*, 22(19):7602, 2022.

[30] M. Oquab, T. Darcet, T. Moutakanni, H. Vo, M. Szafraniec, V. Khalidov, P. Fernandez, D. Haziza, F. Massa, A. El-Nouby, et al. DINOv2: Learning robust visual features without supervision. *arXiv preprint arXiv:2304.07193*, 2023.

[31] P. Palencia, R. Vada, S. Zanet, M. Calvini, A. De Giovanni, G. Gola, and E. Ferroglio. Not just pictures: Utility of camera trapping in the context of african swine fever and wild boar management. *Transboundary and Emerging Diseases*, 2023:1–9, 2023.

[32] K. Papafitsoros, L. Adam, and G. Schofield. A social media-based framework for quantifying temporal changes to wildlife viewing intensity. *Ecological Modelling*, 476:110223, 2023.

[33] J. R. Parham, J. Crall, C. Stewart, T. Berger-Wolf, and D. Rubenstein. Animal population censusing at scale with citizen science and photographic identification. In *2017 AAAI Spring Symposium Series*, 2017.

[34] P. C. Ravoor and T. Sudarshan. Deep learning methods for multi-species animal re-identification and tracking–a survey. *Computer Science Review*, 38:100289, 2020.

[35] V. Renò, G. Dimauro, G. Labate, E. Stella, C. Fanizza, G. Cipriano, R. Carlucci, and R. Maglietta. A SIFT-based software system for the photo-identification of the Risso's dolphin. *Ecological informatics*, 50:95–101, 2019.

[36] J. M. Rowcliffe, J. Field, S. T. Turvey, and C. Carbone. Estimating animal density using camera traps without the need for individual recognition. *Journal of Applied Ecology*, pages 1228–1236, 2008.

[37] S. Schneider, G. W. Taylor, and S. C. Kremer. Similarity learning networks for animal individual re-identification: an ecological perspective. *Mammalian Biology*, pages 1–16, 2022.

[38] S. Schneider, G. W. Taylor, S. Linquist, and S. C. Kremer. Past, present and future approaches using computer vision for animal re-identification from camera trap data. *Methods in Ecology and Evolution*, 10(4):461–470, 2019.

[39] G. Schofield, M. Klaassen, K. Papafitsoros, M. K. S. Lilley, K. A. Katselidis, and G. C. Hays. Long-term photo-id and satellite tracking reveal sex-biased survival linked to movements in an endangered species. *Ecology*, 101(7):e03027, 2020.

[40] A. Shukla, G. Sigh Cheema, P. Gao, S. Onda, D. Anshumaan, S. Anand, R. Farrell, et al. A hybrid approach to tiger re-identification. In *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops*, 2019.

[41] C. Trotter, G. Atkinson, M. Sharpe, K. Richardson, A. S. McGough, N. Wright, B. Burville, and P. Berggren. NDD20: A large-scale few-shot dolphin dataset for coarse and fine-grained categorisation. *arXiv preprint arXiv:2005.13359*, 2020.

[42] B. P. C. Trust. Panthera pardus csv custom export, 2022.

[43] V. Čermák, L. Picek, L. Adam, and K. Papafitsoros. Wildlifedatasets: An open-source toolkit for animal re-identification. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 5953–5963, 2024.

[44] M. Vidal, N. Wolf, B. Rosenberg, B. P. Harris, and A. Mathis. Perspectives on individual animal identification from biology and computer vision. *Integrative and comparative biology*, 61(3):900–916, 2021.

[45] O. Wahltinez and S. J. Wahltinez. An open-source general purpose machine learning framework for individual animal re-identification using few-shot learning. *Methods in Ecology and Evolution*, 15(2):373–387, 2024.

[46] J. Wall, G. Wittemyer, B. Klinkenberg, and I. Douglas-Hamilton. Novel opportunities for wildlife conservation and research with real-time monitoring. *Ecological Applications*, 24(4):593–601, 2014.

[47] L. Wang, R. Ding, Y. Zhai, Q. Zhang, W. Tang, N. Zheng, and G. Hua. Giant panda identification. *IEEE Transactions on Image Processing*, 30:2837–2849, 2021.

[48] C. L. Witham. Automated face recognition of rhesus macaques. *Journal of Neuroscience Methods*, 300:157–165, 2018.

[49] M. Zuerl, R. Dirauf, F. Koeferl, N. Steinlein, J. Sueskind, D. Zanca, I. Brehm, L. v. Fersen, and B. Eskofier. Polarbearvidid: A video-based re-identification benchmark dataset for polar bears. *Animals*, 13(5):801, 2023.

[50] S. Zuffi, A. Kanazawa, T. Berger-Wolf, and M. J. Black. Three-D safari: Learning to estimate zebra pose, shape, and texture from images "In the wild". In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 5359–5368, 2019.